Talk to Your Brain: Artificial Personalized Intelligence for Emotionally Adaptive AI Interactions

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***Abstract*—The process of =making an Artificial intelligent system to replicating human emotional understanding and adapt its responses contextually by tailoring its responses based on individual cognitive and emotional states is called Artificial Personalized Intelligence (API). In this paper, we present a fine-tuned emotionally adaptive AI pipeline capable of generating personalized, human-like responses. Using a custom dataset based on six universal emotions—Sadness, Happiness, Fear, Anger, Surprise, and Disgust—collected from interviews with 30 participants, we have explored two approaches: fine-tuning a LLaMA-3 8B model with the Low-Rank Adapter (LoRA) technique and employing a Retrieval-Augmented Generation (RAG) agent-based framework over the same LLaMA-3 8B model. Emotion classification, prompt engineering, and model fine-tuning were used over the dataset to capture emotional and personalized subtleties in the responses based on different individuals. We conducted a comprehensive analysis, evaluating performance across candidates, emotions, and overall performance, using metrics such as Mean Squared Error (MSE) and Pearson Correlation Coefficient to measure the difference among the outputs from both pipelines and actual human emotional responses. Our results show that prompt engineering combined with LoRA-based fine-tuning significantly enhances the ability to engage in emotionally intelligent and personalized conversations, whereas the RAG-based approach underperformed due to the pretrained LLAMA-3 model's restrictive neutral and adversarial training. This highlights that fine-tuned LLMs are effective in replicating human emotions for personalized intelligence.**

***Index Terms*—Artificial Personalized Intelligence, Generative Artificial Intelligence, Large Language Models, Cognitive Intelligence, Retrieval-Augmented Generation, Fine-tuning**

I. Introduction

In today's rapidly evolving world, there is growing interest among researchers in understanding how the human mind works, interprets complex data, and makes decisions while maintaining individuality in challenging situations. The brain's remarkable ability to handle these tasks has inspired the development of artificial personalized intelligence (API) systems that imitate cognitive processes and provides personalized answers. Such AI systems, equipped with contextual awareness and emotional touch, hold the potential to solve problems, identify trends, and make decisions with a level of personalization and adaptability akin to human intelligence. [1].

Modern generative AI models are highly effective at processing large datasets and generating precise outputs, yet they face significant challenges in replicating human thought processes, particularly in areas such as introspection, emotional nuance, and self-perception. Interoception, which involves reflecting on personal experiences, managing emotions, and regulating intrusive thoughts, is deeply rooted in individual experiences and emotional complexity. AI systems currently struggle to navigate these subtleties, resulting in a disconnect during cognitive and emotional interactions with users [2]. A major limitation of these AI models in achieving personalized intelligence is their inadequate self-reflection capabilities. While some models can recognize basic emotions, they fall short of replicating the depth of self-reflective reasoning that is crucial to human cognition. AI systems are unable to emulate the internal mechanisms of interoception or engage in the complex cognitive dialogues required for effective emotional regulation [3]. Additionally, these models cannot manage intrusive thoughts, which are common across various mental health conditions. This shortcoming is particularly important in therapeutic settings, where self-reflection and resolving emotional conflicts are essential. Current models are not yet sophisticated enough to assist in navigating these emotional challenges [4]. Moreover, the lack of personalization in AI responses further hampers their ability to connect meaningfully with individuals. Most models provide generalized responses that fail to capture the unique emotional complexity of each user's cognitive experiences, making AI interactions seem impersonal and distant, and ultimately reducing their effec-

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tiveness in facilitating the introspective discussions needed for deep self-reflection.

Given these limitations, the development of highly individualized AI systems—personalized AI framework becomes critical. These systems would act as personal cognitive assistants, designed to support individuals in decision-making by providing insights tailored to their unique emotional and cognitive needs. An Artificial Personalized Interface (API) could enable individuals to interact with personalized AI agents that guide decision-making based on their own experiences and specific requirements.

The major contributions of this paper are as follows, offering significant advancements in the field of Artificial personalized intelligence to mimic human emotional reasoning:

1) A comprehensive custom questionnaire was created and 30 candidates were interviewed from diverse backgrounds, based on psychological theories and research on the six universal emotions of Sadness, Happiness, Fear, Anger, Surprise, and Disgust [3]. This custom dataset was utilized both for training the model as well as establishing testing for emotions.
2) The state-of-the-art open-source LLaMA-3 8B model was finetuned on the custom dataset using LoRA. Customized prompts were then used to generate answers that closely resemble human emotional responses from the questionnaire.
3) The same LLaMA-3 8B was also employed with a Retrieval-Augmented Generation (RAG) and agent-based framework, consisting of custom prompts and a context database from the custom dataset, to generate responses that closely resemble human emotional responses from the questionnaire.
4) A Comparison of performance of both pipelines were performed and it was found that the RAG pipeline underperformed over the fine-tuned model, which stems from the pre-trained LLAMA-3's adversarial training for safe and neutral outputs. [5]. This impairs the model's ability to generate personalized emotional responses, thus resulting in lower Pearson Correlation and higher MSE scores for personalized emotional responses for the RAG pipeline.

II. Related Work

The previous research on AI-driven chatbot systems reveals various approaches aimed at improving emotional recognition, empathy generation and cognitive abilities of the models. However, despite such advancements, various challenges related to data noise, personalization, and nuanced understanding of cognitive emotions still exist.

Zhengyi Ma et al. [6] proposed aD HAP model to resolve the issue of personalized chatbot responses, which learns implicit user profiles based on interaction history. Despite generating user-specific and contextually relevant responses, the model however struggles with handling noisy data and faces limitations in performance scalability.

Yanran Li et al. [7] introduced Emma, which rather than simply classifying emotions into predefined categories is designed to grasp users' emotions by focusing on the causes of these emotions. This approach moves beyond traditional emotion analysis models, yet encounters difficulties in fully capturing the complexity of human emotional states by providing generic responses.

Sotolar et al. [8] proposed a method for enhancing empathetic responses in large language models by combining preference optimization with theory-driven datasets. By using emotional labels from Plutchik's wheel of emotions, the pipeline enhances the empathetic capabilities of chatbots, however the system is prone to overfitting during fine-tuning and struggles with stability issues related to random completion selections.

There have been few researchers who have worked on modelling human individuality by tackling personalization. Yuan Li [9] investigated the psychological traits of large language models, examining personality, values, emotions, and motivation. The study highlights discrepancies between the self-identified traits of these models and their actual behaviors, highlighting the challenges involved in aligning AI responses with a user's psychological profile in practical applications.

Huang et al. [10] produced an AI-powered chat application designed to support patients with anxiety and depression. The application enable structured conversations but fell short in delivering dynamic and personalized feedback necessary for genuine reflective dialogue.

Zhou et al. [3] explore emotion-driven AI personalization, focusing on chatbots that detect user emotions and respond empathetically. While these systems can recognize emotional cues, they fail to incorporate individual experiences and often provide generalized, repetitive responses. This study compares fine-tuned open-source LLM model and a RAG-based agent framework using a custom questionnare dataset from 30 participants aiming rather than generating generalized responses, to generate personalized, emotionally aware responses for more human-like AI interactions.

III. Methodology

A. *Data Collection*

To develop an AI system capable of replicating human emotional responses, a custom questionnaire was designed based on psychological theories and research on six universal emotions: Sadness, Happiness, Fear, Anger, Surprise, and Disgust [3]. The questionnaire targeted key emotional and cognitive processes, including emotional triggers, coping strategies, decision-making, and emotional regulation.

The questionnaire was divided into different sections based on key emotional and cognitive processes, ensuring a comprehensive understanding of emotional triggers and coping strategies thus bringing out one's individuality:

1) Demographic Information: Contextualizes their emotional responses (e.g., age, profession).
2) Mental State: Explores frequency of emotions like stress and coping methods.

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3) Emotional Reactions: Investigates responses to success, criticism, and failure.
4) Problem Solving: Assesses reflective thinking and decision-making.
5) Coping with Change: Examines reactions to sudden changes.
6) Emotional Regulation: Explores how emotions like anger and fear are managed.
7) Specific Scenarios: Elicits emotional responses through hypothetical situations.

Interviews were conducted with 30 students from diverse backgrounds in a standardized, soundproof environment to minimize bias and ensure data integrity. Each 20 to 30-minute session explored a range of emotions, and the audio recordings were transcribed to provide detailed insights into participants' reactions. This data served as the foundation for training as well as testing the AI frameworks to simulate emotionally intelligent and contextually relevant interactions.

B. *Data Processing*

The data processing pipeline involved several key stages, starting with audio transcription using the Deepgram Nova-2 model, selected for its high accuracy and speed. Then processes such as diarization to distinguish between speakers and smart formatting for automated punctuation and capitalization were performed. Diarization ensured clear separation between interviewer and interviewee, while smart formatting enhanced readability with minimal manual editing.

After transcription, question-answer pairs were classified into scenarios based on dialogue structure. Scenario 0 included simple Q&A exchanges, while Scenario 1 featured deeper emotional and cognitive probing. This classification helped organize the data for training the AI model to distinguish between basic and complex interactions.

The text was further processed by consolidating split queries and responses into coherent paragraphs, removing irrelevant noise, and ensuring consistent formatting. Finally, the embeddings of phrases from the transcript were clustered using cosine similarity to group semantically similar questions, revealing emotional trends and discourse patterns. This structured dataset allowed the AI model to better understand emotional signals and engage in contextually relevant, emotionally intelligent conversations.

C. *Model Fine-Tuning and Emotional Response Engineering*

The open-source LLaMA-3 8B model was fine-tuned on the processed dataset using the Low-Rank Adapter (LoRA) technique. The fine-tuning process was done using Hugging Face's 'transformers' library, specifically the 'AutoModelForCausalLM' class, along with a 4-bit quantization method to maximize memory efficiency without compromising performance. The dataset was first converted into a conversational format and tokenized using the LLaMA-3 tokenizer to ensure input consistency. The model was then loaded in 4-bit precision, utilizing NF4 quantization through the `BitsAndBytesConfig` object to optimize memory usage during training.

For the fine-tuning, the attention layers were adjusted using the LoRA (Low-Rank Adaptation) approach, with key parameters set to Rank 16, LoRA Alpha 32, and LoRA Dropout 0.05, focusing on Causal Language Modeling. The training configuration included a learning rate of 5e-5, a batch size of 2, gradient accumulation steps of 1, and a maximum sequence length of 512. The Paged AdamW optimizer was employed in 32-bit precision to ensure optimal performance. The training process was executed using Hugging Face's 'transformers' library, with 'wandb' used for tracking and monitoring the training metrics. After one epoch of training, the model was saved for further evaluation. The use of LoRA and quantization allowed the model to adapt effectively to the dataset while maintaining computational efficiency.

To enhance the model's ability to generate emotionally adaptive responses, prompt engineering techniques were employed. These techniques enabled the model to generate emotions specific personalized responses such as anger, happiness, sadness, and neutrality. The process began with emotion classification, where input text was analyzed using the pre-trained Hugging Face 'emotion-english-distilroberta-base' model via the 'pipeline'. Based on the identified emotion, a corresponding prompt template was selected to match the emotional tone. After the custom prompts were formatted, A proper token limit was set for each response to maintain coherent conversational outputs. The architecture diagram for the finetuned LLM architecture is shown in 1.

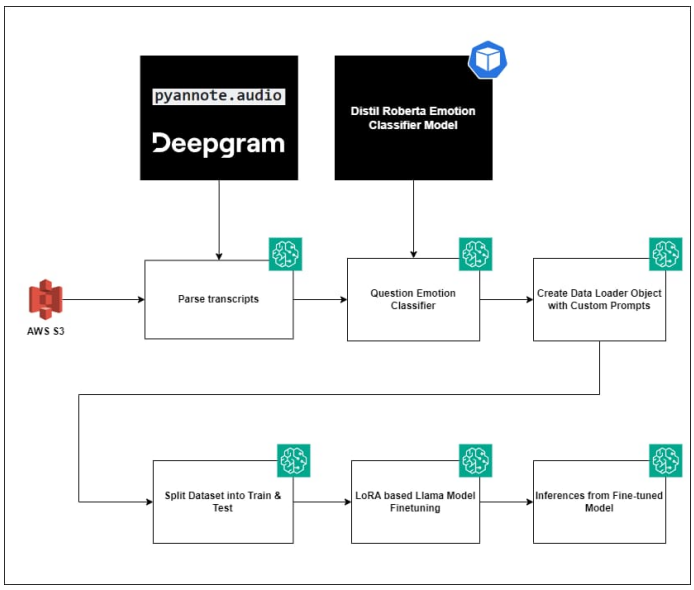

Fig. 1. Architecture diagram for end-end pipeline using fine-tuned LLM

D. *RAG based Agent Framework*

For generating enhanced responses, the answers are produced by integrating emotion recognition with context-based

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database retrieval. The framework follows a structured sequence of steps, ensuring that the responses are not only accurate but also emotionally relevant.

Emotion Recognition: The initial step involves recognizing the emotion conveyed by the input question. This is achieved using a bidding mechanism powered by the open-source LLaMA-3 8B model. The LLM analyzes the input to determine the emotional tone, which can vary from anger or joy to sadness or fear. This emotional classification is crucial, as it directs the subsequent stages of response generation.

Once the emotion of the input is identified, the framework retrieves relevant content through two primary components:

1) **Answer Retrieval:** The system fetches potential answers from a pre-existing knowledge base designed from the custom questionnaire dataset that aligns with the topic of the question.
2) **Chat History Retrieval:** Additionally, the system retrieves previous interactions or chat history related to the current question. This ensures coherence and contextual relevance in the final response.

With the relevant content retrieved and the emotion identified, the final answer is generated by evaluating both the retrieved answers and the chat history to determine the most suitable response. The response is then tailored based on the recognized emotion. For instance, if sadness is detected, the system crafts a response to offer empathy or comfort.

By integrating emotion recognition with context-based content retrieval, this approach ensures that responses are both contextually relevant and emotionally attuned. This combination significantly enhances the user experience by addressing the user's informational needs while also providing an emotionally appropriate response.

IV. Results & Discussion

A. *Evaluation Metrics Used*

1) Mean Squared Error (MSE): Mean Squared Error (MSE) is a commonly used metric to evaluate the performance of regression models. It measures the average squared difference between the predicted values and the actual values. Formally, for a dataset with n samples, let y_i represent the actual values and $\hat{y}_i$ represent the predicted values. The MSE is defined as:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{1}$$

2) Pearson Correlation Coefficient: The Pearson Correlation Coefficient (r) measures the linear correlation between the predicted and actual emotional values across both models. It is computed as:

$$r = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}\sqrt{\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}} \tag{2}$$

B. *Hyperparameters and Configuration*

As detailed in Table I and Table II, we utilized a specific LoRA configuration and defined training arguments to optimize the model's performance. The LoRA settings focus on causal language modeling, while the training parameters ensure efficient model training and evaluation.

TABLE I
LoRA Configuration

Parameter	Value
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Bias setting	None
Task type	Causal Language Modeling (CAUSAL_LM)

TABLE II
Training Arguments

Parameter	Value
Training batch size (per device)	2
Evaluation batch size (per device)	2
Gradient accumulation steps	1
Optimizer	AdamW
Number of training epochs	10
Evaluation strategy	Steps
Evaluation steps	10
Logging steps	10
Warmup steps	0
Learning rate	5e-5
Mixed precision (FP16)	Enabled
BF16 precision	Disabled
Sequence length grouping	Enabled

C. *Candidate-wise analysis*

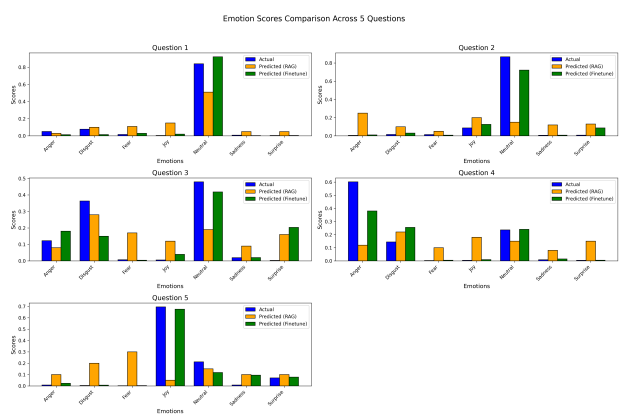

Fig. 2. Actual v/s predicted emotion scores for Candidate 5

The emotion score comparison across five questions for Candidate 5 present in Fig 2 shows that the fine-tuned model closely aligns with the actual emotion scores, particularly in emotions like sadness, fear, and joy, consistently outperforming the RAG model. The RAG model, however, frequently underperforms, especially in emotions like anger and disgust,

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where its predictions are significantly lower than the actual values. Additionally, the RAG model tends to predict neutral responses more often, which affects its ability to capture the emotional nuances of other categories. This underperformance can be attributed to the pre-trained LLAMA-3 model's adversarial training for neutral and safe outputs. While this training serves a valuable purpose in ensuring non-harmful outputs, it inadvertently impairs the model's ability to generate accurate and nuanced emotional responses.

D. *Emotion Analysis:*

TABLE III
PEARSON CORRELATION AND MSE FOR EMOTIONS (RAG VS. FINETUNING)

Emotion	Pearson Correlation		MSE	
	RAG	Finetuning	RAG	Finetuning
Anger	0.783	0.850	0.0568	0.0452
Disgust	0.821	0.890	0.0284	0.0210
Fear	0.691	0.740	0.0107	0.0086
Joy	0.705	0.780	0.0051	0.0042
Neutral	0.842	0.900	0.0476	0.0365
Sadness	0.668	0.730	0.0099	0.0078
Surprise	0.619	0.690	0.0043	0.0037

From table III it can be inferred that the fine-tuned LLAMA3 model consistently outperforms RAG in both metrics across all emotions. For instance, in the case of "Anger," Finetuning achieves a Pearson Correlation of 0.850, notably higher than RAG's 0.783, and a lower MSE (0.0452 vs. 0.0568). This trend is observed for all emotions, with significant performance improvements seen in "Disgust" and "Neutral." The Finetuning approach demonstrates stronger alignment with the actual emotional labels and makes more accurate predictions, as evidenced by its higher Pearson Correlation scores and consistently lower MSE values compared to the RAG model. The RAG model struggles to capture emotional subtleties due to its dversarial training for neutral and safe outputs, leading to lower Pearson Correlation and higher MSE scores. In contrast, the Finetuning model, being directly optimized for emotion recognition, bypasses the limitation and excels in generating accurate emotional responses.

E. *Overall Performance*

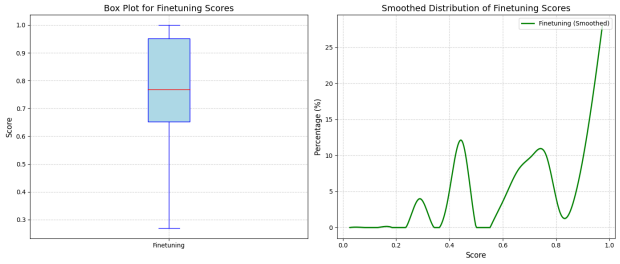

Fig. 3. Distribution plot for average cosine similarity scores for the candidates

The box plot and smoothed distribution graph presented in Fig. 3 for the fine-tuned LLAMA-3 8B model's scores highlight the model's strong performance in predicting emotional responses. With a median score of 0.77, the box plot suggests that the majority of predictions align closely with actual human emotions, with scores predominantly ranging from 0.7 to 0.9. While their is a visible smaller peak at 0.4 in the smoothed distribution graph, indicating occasional challenges in capturing subtle emotional nuances, these instances are relatively infrequent. And it can be seen that a significant percentage of questions per candidate achieved high cosine similarity, with a bigger peak around 0.9, confirming the model's overall effectiveness. The average cosine similarity score remains strong at 0.76.

Upon evaluating individual candidates, Candidate 5 emerged with the highest cosine similarity, reflecting a strong alignment between the model's predictions and actual emotional states. Out of 150 questions, 18% of the predictions exhibited cosine similarity scores between 70-80%, while an impressive 44% surpassed 80%. Additionally, 22% fell within the 50-70% range, and only 16% scored below 50%. Pearson correlation analysis for emotions such as Disgust and Fear revealed that the model achieved an accuracy of 70-80% when evaluating emotions collectively, rather than on a response-by-response basis. These results underscore the model's reliability in predicting emotions for both individuals and larger groups. In contrast, Candidate 6 presented challenges, with the model showing prediction accuracies ranging from 60% to 80% for certain emotional states

V. CONCLUSION AND FUTURE WORKS

Artificial Personalized Intelligence seeks to imitate human emotional understanding and adapt AI interactions based on individuality of the human mind over emotional and cognitive states. In this paper, we fine-tuned a LLaMA-3 8B model using a Low-Rank Adapter (LoRA) and implemented a Retrieval-Augmented Generation (RAG) agent-based framework, evaluating their ability to replicate personalized emotional responses based on a custom questionnaire dataset of six universal emotions—collected from interviews with 30 participants. The results indicate that the LoRA fine-tuning significantly enhanced the AI's emotional prediction capabilities, with scores closely aligning with human responses, particularly in emotions like sadness, fear, and joy. The LoRA model consistently achieved higher Pearson Correlation and lower Mean Squared Error, indicating better alignment with human emotional reasoning across multiple metrics. In contrast, the RAG model underperformed, primarily due to its adversarial training for neutral and safe responses, which resulted in diminished emotional sensitivity. The retrieval mechanism further compounded this by pulling neutral, less emotionally rich content, leading to lower predictive accuracy, especially in nuanced emotions like joy and disgust. Future work will focus on addressing these limitations by integrating more emotionally rich external data and exploring multi-modal data to improve real-time adaptability and emotional depth in personalized AI interactions.

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Furthermore we plan to experiment with larger models such as LLaMA 40B, DBRX, GPT-4 and Claude for generating such responses, thus providing a more comprehensive evaluation of the approach.

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